

Intergenerational Housing Misallocation and Fertility

Short model note

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I set up a simplified version of the model. There are two types of agents, young and old, children require space, and a combination of a down payment friction and a tax friction generates intergenerational misallocation which impacts fertility. Results are preliminary, and proofs are omitted. The model draws inspiration from Coven et al. (2025). In this simplified version, fertility is continuous.

1 Environment

The economy is populated by overlapping generations of young and old households. A young household has type $i = (y_i, a_i)$: income and wealth at entry. Wealth is inherited from the previous old generation¹. A household decides whether to enter the economy; inside, she chooses a tenure (rent or own), a house size, completed fertility, and savings for old age. Old households, indexed by j , are of two kinds. Old incumbents, last period's young owners, who enter old age with financial wealth, the house they bought, and a possible accrued gain, choose how much of the house to keep and how large an estate to leave, then exit. The other type is old renters, last period's young renter entrants, who rent in old age and carry no wedge. Nonentrants stay outside the market in both periods. Each period a mass $\bar{M}$ of potential young households enters the pool: the maturing children of the previous cohort plus a renewal inflow. Types are distributed according to an exogenous G_Y .

Preferences. Households have preferences over consumption, housing, and children. Children raise the household's consumption and housing needs: each child claims $\chi > 0$ units of consumption and $\kappa > 0$ units of space, so with consumption spending c_i , housing h_i , and fertility n_i ,

$$u_i^Y = \log(c_i - \chi n_i) + \alpha \log(h_i - \kappa n_i) + \vartheta \log n_i.$$

Old households value consumption, housing, and the bequest $e_j \geq 0$ they leave at exit:

$$u^O = \log c_j^O + \gamma \log h_j^O + \mathcal{B}(e_j),$$

with $\mathcal{B}$ a warm-glow function, increasing and concave, possibly zero, common to all old households. The exiting cohort's estates are what the arriving cohort inherits.

Housing. There is a fixed stock $\bar{H}$ of divisible floorspace, held by old incumbents (their carried houses) and by passive landlords (the rest): buyers purchase from either, renters occupy landlord

¹Distributed randomly, without a tie to bequests and inheritants. It can be distributed unequally, however.

space, and rents and sale proceeds are transfers. Renting and owning are occupancy modes with different size limits, the largest sizes available to rent and to own,

$$h \leq \bar{h}^R \text{ (rent)}, \quad h \leq \bar{h}^O \text{ (own)}, \quad \bar{h}^R < \bar{h}^O,$$

Family-sized houses are mostly available to buy, not to rent. All floorspace trades in one market: the asset price is P , and no-arbitrage ties the rental rate to it,

$$q = \rho P, \quad \rho = r + \delta + \tau^P.$$

Given q , a higher property tax lowers P and with it the down payment on any given house. Lowercase h is an occupancy choice; capital H marks housing quantities not chosen today: the stock, the caps, carried housing.

Financing. A buyer finances at most a share $\phi \in (0, 1)$ of the purchase, so she posts a down payment $(1 - \phi)Ph$ at purchase, out of her liquid wealth. Households can save in an exogenously provided bond, but renters cannot borrow. The interest rate r is exogenous.

Taxes. Two taxes appear: the property tax τ^P , part of the user cost above, and the capital-gains tax. Selling a house triggers a tax: the seller pays the rate τ^g on the difference between the sale price and what she paid, in dollars. Dying instead passes the house with a stepped-up basis - the heir's purchase price is reset to the current value -so the gain is never taxed².

Where do gains come from when the real price never moves? From inflation. Dollar prices and incomes all grow at rate ϖ , so relative prices and real choices are unaffected; but the tax form compares dollars from different dates, and a house bought one period ago shows a paper gain even though its real value is unchanged. The tax on that paper gain is real,

$$\bar{\ell} = \tau^g \frac{P \varpi}{1 + \varpi}$$

per unit sold, and this is the model's only contact with inflation. Every cohort carries the same gain, so every incumbent's unit bears $\bar{\ell}$ if sold while alive.³ An estate tax, by contrast, would fall on the transfer in any form and leave the sell-or-keep margin undistorted; the gains tax distorts it precisely because death forgives it. Everything is deterministic: a buyer knows the tax her future self will face⁴.

2 Households

Conditional on entry, household i chooses a tenure mode $m \in \{R, O\}$, housing, fertility, consumption and savings; old age is discounted at $\beta \in (0, 1]$. For $m \in \{R, O\}$,

$$W_i^m = \max_{c_i, h_i, n_i, a'_i} \log(c_i - \chi n_i) + \alpha \log(h_i - \kappa n_i) + \vartheta \log n_i \\ + \beta [\mathbf{1}\{m = R\} V^R(a_{j'}) + \mathbf{1}\{m = O\} V^O(a_{j'}, h_i)]$$

²This was the essence of the point suggested by Simon. I will go a bit deeper into it in the next weeks, but it is one of the two key margins (the other being financial constraints). The idea is then: agents have an incentive to hold on to the house, because this way neither they nor their heirs pay taxes.

³The statutory exclusion, which exempts gains below a dollar threshold, is an institutional detail I suppress; with it, small holdings escape the tax and lock-in sorts by size.

⁴So I am resorting to a trick here to have a capital gain in a stationary model. COVEN ET AL have a real growth term there, but they don't have my problem with fertility. More thought is required on this.

subject to

$$c_i + qh_i + a'_i = y_i + a_i, \quad h_i \leq \bar{h}^m, \quad a'_i \geq 0, \\ \mathbf{1}\{m = O\} (1 - \phi)Ph_i \leq a_i.$$

Children have a market price: each child claims χ goods and κ units of space, and space rents at q , so providing for a child costs $\chi + \kappa q$ per period. A useful benchmark is equal scaling, $\chi = \kappa q / \alpha$: the child splits her cost between space and goods in the proportion $\kappa q / \chi = \alpha$, the same proportion in which her parents split their own spending.

The old state is generated by the household's own choices: financial wealth $a_{j'} = (1 + r) a'_i - \mathbf{1}\{m = O\} qh_i$ —real wealth is conserved, the nominal gain is not wealth. Nothing is random; V^O and V^R are the old values defined next.

The old incumbent enters with financial wealth a_j , housing H_j^0 , and an accrued gain; old households have no income. Each unit she holds faces one choice: sell now and pay the tax, or keep it until death, when the tax is forgiven. She solves

$$V^O(a_j, H_j^0) = \max_{c_j^O, h_j^O, e_j} u^O(c_j^O, h_j^O, e_j) \\ \text{subject to } c_j^O + e_j + q h_j^O = a_j + q H_j^0 - \bar{\ell} (H_j^0 - h_j^O), \quad 0 \leq h_j^O \leq H_j^0.$$

Selling a unit nets $q - \bar{\ell}$; the forgiveness at death is a subsidy to staying put, a transfer rather than a resource saving. An old renter has no carried housing and no tax:

$$V^R(a_j) = \max_{c_j^O, h_j^O, e_j} u^O(c_j^O, h_j^O, e_j) \quad \text{subject to } c_j^O + e_j + q h_j^O = a_j, \quad 0 < h_j^O \leq \bar{h}^R,$$

and at an interior choice her marginal value of space is exactly q .

Using $P = q / \rho$, owner housing is bounded by Using $P = q / \rho$, owner housing is bounded by

$$H_i^O = \min \left\{ \bar{h}^O, \frac{a_i \rho}{(1 - \phi) q} \right\}, \quad H_i^R = \bar{h}^R.$$

H_i^m is the effective family-housing cap: a renter is capped by the stock, a buyer by the size limit or by collateral, whichever is shorter. Tenure is $W_i^Y = \max\{W_i^R, W_i^O\}$. Entry compares the inside value with an outside option $\bar{W}^E$ under extreme-value shocks of scale κ_E :

$$\pi_i^E = \frac{\exp\{W_i^Y / \kappa_E\}}{\exp\{W_i^Y / \kappa_E\} + \exp\{\bar{W}^E / \kappa_E\}}$$

is the probability that household i enters.

3 Equilibrium

Aggregate demand adds young households, old incumbents, and old renters, and the single market clears:

$$\bar{M} \int \pi_i^E h_i dG_Y(i) + \int h_j^O dF_O(j) + \int h_j^O dF_R(j) = \bar{H},$$

where F_O and F_R are the measures of old incumbents and old renters. A stationary equilibrium consists of a user cost q (and $P = q / \rho$), young policies and entry probabilities, old policies, and stationary objects $(\bar{M}, G_Y, F_O, F_R)$ such that:

1. the policies solve the household problems above at q ;
2. the floorspace market clears, as displayed;
3. the population reproduces itself: young owners become the old incumbents, young renters the old renters, and births plus the renewal inflow recreate the cohort;
4. fiscal and financial accounts close with lump-sum transfers.

Nothing in the clearing condition asks whether the marginal family-sized unit is held by the household that values it most; that is the next section's question.

4 Equilibrium policies

How much a constraint binds can be read off the household's bundle: the consumption, housing, and fertility she chooses. The case problem's Lagrangian is

$$\begin{aligned}\mathcal{L} = & u_i^Y + \beta V_{j'} + \lambda_i (y_i + a_i - c_i - qh_i - a'_i) \\ & + \mu_i (a_i - (1 - \phi)Ph_i) + \theta_i (\bar{h}^m - h_i) + \xi_i a'_i,\end{aligned}$$

with $\mu_i = 0$ in the rental mode and $\xi_i \geq 0$ the multiplier on the no-borrowing bound $a'_i \geq 0$ —for a renter, the constraint that she cannot borrow at all; every condition in this section is a derivative of $\mathcal{L}$.

The first-order condition for saving sets the marginal value of resources today against the return tomorrow:

$$\frac{1}{c_i - \chi n_i} = \beta(1 + r) \frac{1}{c_{j'}^O} + \xi_i, \quad \xi_i a'_i = 0,$$

which for savers ($\xi_i = 0$), under logarithmic old utility (warm glow $b \log e$ included) and interior old choices, collapses to a savings rule:

$$a'_i = \beta(1 + \gamma + b)(c_i - \chi n_i) + \mathbf{1}\{m = O\} \frac{\bar{\ell} h_i}{1 + r},$$

with γ the old housing weight and b the warm-glow weight: the household saves old age's claim on adult consumption—write $k \equiv \beta(1 + \gamma + b)$ for that load—plus her discounted future tax bill, so anticipating the lock-in tax is itself a savings motive. By the same conversion, a buyer pays an effective price $q^O = q + \bar{\ell}/(1 + r)$ per unit of space—the market price plus her own discounted lock-in tax—while a renter pays q ; write q^m for the case's effective price. A unit of adult consumption commits $1 + k$ units of resources, one spent now and k carried.

When no cap binds, the allocation follows in one line: with lifetime resources $w_i = y_i + a_i$,

$$c_i - \chi n_i = \frac{w_i}{1 + \alpha + \vartheta + k}, \quad n_i = \frac{\vartheta w_i}{(1 + \alpha + \vartheta + k)(\chi + \kappa q^m)}, \quad h_i = \frac{\alpha w_i}{(1 + \alpha + \vartheta + k) q^m} + \kappa n_i,$$

and savings follow the rule. When the cap binds, $h_i = H_i^m$, fertility solves the condition displayed in the last section, and the budget gives the rest.

Dividing the housing derivative of $\mathcal{L}$ by the consumption derivative, $\lambda_i = 1/(c_i - \chi n_i)$, prices every multiplier in goods. For a renter this gives

$$\frac{\alpha(c_i^R - \chi n_i^R)}{h_i^R - \kappa n_i^R} = q + \zeta_i^R, \quad \zeta_i^R = \frac{\theta_i}{\lambda_i} \geq 0 :$$

the left side is u_h/u_c , her marginal rate of substitution between space and consumption—the goods she would give up for one more unit of space, holding utility fixed—and it exceeds the rent exactly by the bite of the rental size limit, computable from her observed bundle. For a young owner the same two derivatives give

$$\frac{\alpha(c_i^O - \chi n_i^O)}{h_i^O - \kappa n_i^O} = q^O + \zeta_i^O, \quad \zeta_i^O = \underbrace{\frac{\mu_i(1-\phi)P}{\lambda_i}}_{\zeta_i^{O,F}, \text{ down payment}} + \underbrace{\frac{\theta_i}{\lambda_i}}_{\text{size limit}} :$$

her marginal value of space exceeds her effective price by the bite of whichever access constraints bind—the down payment (μ_i) or the owner size limit (θ_i). When the size limit is slack, the bundle identifies the financing bite alone: $\zeta_i^{O,F}$ is the observed ratio minus q^O . For an old incumbent at an interior retention choice,

$$\frac{\gamma c_j^O}{h_j^O} = q - \bar{\ell} :$$

she keeps marginal space she values below what the market would pay, because selling it would realize the taxable gain. Old renters carry no such wedge: their marginal value is q .

The gaps price children. Multiplying the fertility first-order condition by consumption gives, in either tenure,

$$\frac{\vartheta(c_i^m - \chi n_i^m)}{n_i^m} = \chi + \kappa(q^m + \zeta_i^m),$$

with q^m the case's effective price. A child's space requirement is priced at the household's own shadow value of space. Children get more expensive exactly where family-sized space is hard to reach, with market prices unchanged.

5 Efficiency

The planner faces the economy's real constraints and nothing else: the stock, the size limits, goods feasibility, and the domain conditions. It does not face household budgets, since allocations are supported by lump-sum transfers, and it does not face the financing inequalities, which restrict market purchases rather than occupancy. Let $r_i^F \geq 0$ be the marginal real cost of implementing one more unit of owner space for a financing-constrained household: zero if the planner can reassign directly, positive if it must work through costly instruments, infinite if the constraint is a primitive nothing relaxes.

Proposition 1. *Hold the stationary cross-section, entry, tenure assignments, and savings fixed. The planner's surplus from moving one unit of space from an old incumbent at an interior retention choice to a young owner whose owner size limit is slack and whose financing constraints bind is*

$$\left(q^O + \zeta_i^{O,F} - r_i^F\right) - (q - \bar{\ell}) = \zeta_i^{O,F} + \frac{\bar{\ell}}{1+r} + \bar{\ell} - r_i^F.$$

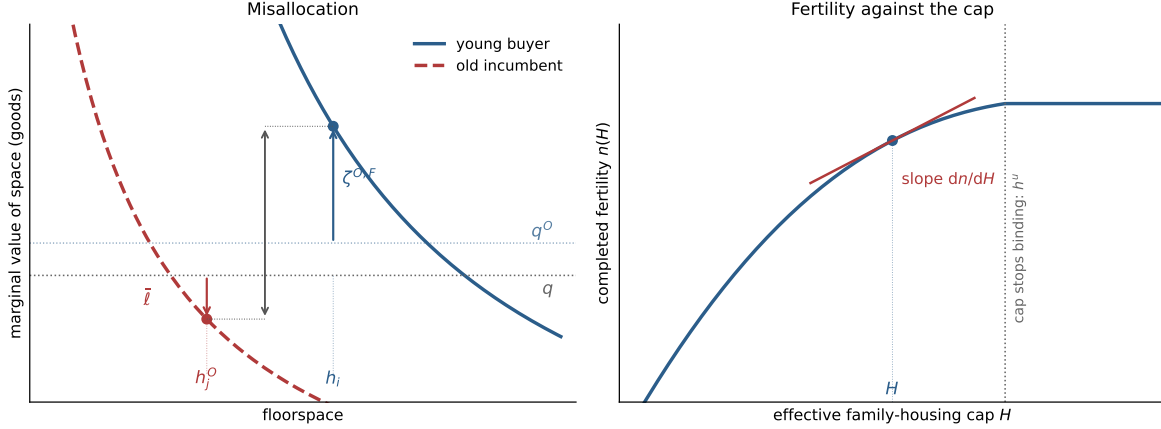


Figure 1: Misallocation and the fertility margin. Left: the collateral-capped buyer values marginal space at $q^O + \zeta^{O,F}$, above the market price; the locked-in incumbent values it at $q - \bar{l}$, below; the distance between the two points is the recoverable surplus. Right: completed fertility against the effective family-housing cap, with slope dn/dH at the equilibrium cap; the cap stops binding at h^u .

The competitive allocation solves the planner's conditional problem only if no feasible pair has positive surplus. Conversely, if all financing gaps and levies are zero, the allocation satisfies the planner's conditional first-order conditions.

To prove it, move dh from the incumbent to the buyer and hold both at their prior utility with lump-sum transfers; the residual $(\zeta_i^{O,F} + \bar{l}/(1+r) + \bar{l} - r_i^F) dh$ in goods is left over, a Pareto improvement under any welfare weights; both tax terms are transfers the planner recovers. Two cautions. Real size limits are not market failures: a planner facing the same stock is pinned with the renter at $\bar{h}^R$. And a positive financing gap is an inefficiency only when an instrument can reach it at real cost below the value gap.

The two margins in one picture. Figure 1 draws a solved stationary case as theory. On the left, the buyer's marginal value of space falls along her constrained path and sits above her effective price by the financing gap $\zeta^{O,F}$ at her cap, while the incumbent's sits below the market price by her tax $\bar{l}$; moving one unit between them frees $\zeta^{O,F} + \bar{l}/(1+r) + \bar{l}$ in goods. On the right, completed fertility rises with the cap at rate dn/dH until the cap stops binding at h^u .

6 Fertility and policy

Fix a tenure case, let H be the effective cap, write $w = y + a$ for lifetime resources, and let q^m be the case's effective price. Concentrating the savings rule into the budget, a binding cap makes fertility solve $\vartheta/n = (1+k)\chi/(w - q^m H - \chi n) + \alpha\kappa/(H - \kappa n)$, and

$$\frac{dn}{dH} = \frac{\frac{\alpha\kappa}{(H - \kappa n)^2} - \frac{(1+k)\chi q^m}{(w - q^m H - \chi n)^2}}{\frac{\vartheta}{n^2} + \frac{(1+k)\chi^2}{(w - q^m H - \chi n)^2} + \frac{\alpha\kappa^2}{(H - \kappa n)^2}},$$

which is positive for every binding cap if and only if $\chi \leq (1 + k) \kappa q^m / \alpha$: relaxing access raises fertility whenever the child's consumption requirement does not exceed its space requirement priced at the lifetime cost of adult consumption. The savings margin loosens the condition because forced housing spending now crowds out saving alongside consumption.

At fixed prices, a small policy z moves aggregate fertility through the constrained:

$$\left. \frac{dN}{dz} \right|_q = \bar{M} \sum_m \int_{\mathcal{E}_m(z)} \left[\pi_i^E \Psi_i^m + n_i^m \frac{\pi_i^E (1 - \pi_i^E)}{\kappa_E} \Theta_i^m \right] \mathcal{P}_i^m(z) dG_Y(i),$$

where $\mathcal{E}_m(z)$ is the set of households whose cap binds in case m , $\mathcal{P}_i^m = \partial H_i^m / \partial z$ is the pass-through of the policy to the binding component of the cap, $\Psi_i^m = dn/dH$ is the fertility response, and $\Theta_i^m = \zeta_i^m / (c_i^m - \chi n_i^m)$ drives the entry term. A policy aimed at a slack constraint moves nothing: a down-payment grant does nothing for a household pinned by the rental size limit. A transfer of spendable resources also moves fertility through the budget at a fixed cap; the formula isolates the access channel. At $\chi = (1 + k) \kappa q^m / \alpha$, equal scaling at lifetime prices, Ψ_i^m is proportional to $\zeta_i^m / (c_i^m - \chi n_i^m)$, so the same gap that measures misallocation predicts whose fertility responds.

Because tenure is a discrete choice, a policy that shifts $W_i^O - W_i^R$ also moves households across the tenure boundary. With no gains tax both tenures price space at the same q , an indifferent household has the same allocation in both, and the switching jump is exactly zero. With the tax, owning is the dear way to occupy space, and it also hands the owner case a constant old-age subsidy—occupying retained space at $q - \bar{\ell}$ —that shifts who sits at the boundary without moving any bundle. The long version prices the tension: the switcher into ownership upsizes when the wedge dominates the subsidy in resources, and she has more children when the space force dominates the subsidy's level shift; at the example's parameters both hold with room, and the boundary household upsizes from 0.15 to 0.20 and adds 0.007 children. The long version of the note states and proves this, together with the stationary demographic closure and the full equilibrium definition.

References

Joshua Coven, Sebastian Golder, Arpit Gupta, and Abdoulaye Ndiaye. Property taxes and housing allocation under financial constraints. Working paper, June 2025, 2025.